

# LÁZARO GÓMEZ VITOLO

AI & Robotics Developer | Python | Machine Learning | Computer Vision | Generative AI  
Argentina • lazarogvitolo@gmail.com • GitHub: Lazaro549 • Portfolio: lazaro549.github.io/Portfolio/

## PROFILE

AI-focused software developer building practical systems across generative AI, evaluation, RAG, computer vision, robotics interfaces, and developer tooling. Strong emphasis on reproducible experiments, measurable results, modular architecture, and engineering workflows that turn AI capabilities into usable software. Public work spans AWS/Bedrock, Anthropic/Claude, MCP, LLM evaluation, and browser-based robotics perception.

## CORE TECHNICAL SKILLS

|                 |                                                                                                                    |
|-----------------|--------------------------------------------------------------------------------------------------------------------|
| AI / ML         | LLM evaluation, LLM-as-a-judge, RAG, embeddings, benchmarking, prompt engineering, agents, ML workflows            |
| Robotics / CV   | Object detection, tracking, real-time perception, telemetry concepts, computer vision pipelines, browser inference |
| Engineering     | Python, TypeScript, JavaScript, FastAPI, React, Vite, Streamlit, REST APIs, SQLite, Docker, GitHub Actions, pytest |
| Cloud / AI APIs | Amazon Bedrock, AWS Lambda, AWS SAM, Anthropic Claude API, MCP, OpenAI-compatible workflows                        |

## SELECTED PROJECTS

### AI Evaluation Platform

Stack: Python, FastAPI, React, TypeScript, SQLite, Docker, GitHub Actions

Production-oriented evaluation platform for LLM applications. Implements deterministic and LLM-as-a-judge metrics, batch evaluation, provider abstraction for Mock/OpenAI/Bedrock, run comparison, reports, REST API, dashboard, and CI without API keys.

### AI Engineering Lab

Stack: TypeScript, Vite, React-style component architecture, GitHub Pages

Technical hub connecting build, evaluate, benchmark, improve, and ship cycles. Enforces evidence integrity by distinguishing measured, demo, and empty results, with tests preventing fabricated metrics or conclusions.

### AWS AI Generative / RAG Benchmark

Stack: AWS Bedrock, FAISS, Python, RAG, AWS Lambda, SAM

End-to-end AWS generative AI repository with chatbot, RAG pipeline, serverless API, and evaluation framework. Controlled retrieval experiment compared chunk size, overlap, and top-k across a 6-document corpus and 18-question dataset; the balanced configuration improved recall from 0.250 to 0.667 while reducing mean retrieval latency.

### Robot Vision System

Stack: TensorFlow.js, COCO-SSD, JavaScript, Vitest

Browser-based perception dashboard using real webcam/video input, object detection, confidence filtering, centroid tracking, inference latency, and FPS metrics. Robot battery, odometry, and network telemetry are explicitly simulated rather than overstated.

### Claude Code / MCP Engineering Workflows

Stack: Claude Code, TypeScript, MCP, GitHub Actions

Built reproducible examples around Claude Code skills, subagents, hooks, and CI, plus an advanced MCP server covering sampling, elicitation, roots, pagination, structured output, Streamable HTTP, resumability, and OAuth 2.1 + PKCE.

## CERTIFICATIONS & LEARNING

Anthropic Academy: Claude Code in Action      Anthropic Academy: Model Context Protocol: Advanced